

KV cache quantization: a technical survey from 2-bit asymmetry to NVFP4

The KV cache — not model weights — is now the dominant memory and bandwidth bottleneck in long-context LLM inference, and quantization has progressed from FP16 to production-grade 4-bit and research-grade sub-2-bit regimes in under 30 months. At Llama-2-70B with 128 K context and batch 32, the FP16 KV cache consumes **1.25 TiB** — nine times the model weights — and every decode step must stream all of it from HBM. Consequently the field has converged on a small set of robust design principles: quantize keys per-channel and values per-token; either quantize keys *before* RoPE or eliminate post-RoPE outliers via Hadamard/polar rotations; reserve FP16 for attention sinks and a sliding residual window; and co-design bit-widths with Tensor-Core-native block formats (FP8 E4M3, NVFP4, MXFP4). This survey traces the algorithmic lineage from SmoothQuant (2022) through KIVI/KVQuant (2024) and QuaRot/SpinQuant (late-2024) to 2025-2026 sub-2-bit vector-quantization schemes (CommVQ, XQuant, H1B-KV, TurboQuant), then analyzes how each interacts with FlashAttention-3, PagedAttention, and the NVFP4 Tensor Cores on Blackwell. It closes with the open problem set: sub-2-bit robustness under long-CoT reasoning, MLA × quantization, and the hardware-software contract for CXL/PIM-resident quantized KV.

1. The memory-bound bottleneck and why keys and values are not the same tensor

Autoregressive decoding reduces every attention layer to a GEMV against the cached (K, V) history. Under the roofline model $P_{\text{attainable}} = \min(P_{\text{peak}}, I \cdot B_{\text{HBM}})$, decode attention has arithmetic intensity $I \approx O(1)$ FLOP/byte — orders of magnitude below the A100's ridge of ~156 FLOP/B. **Decode time-per-output-token (TPOT) is therefore set by bytes moved, not FLOPs,** (Edge AI and Vision Alliance) and compressing KV is the single highest-leverage optimization available.

The exact FP16 KV footprint is

$$M_{KV} = 2 \cdot B \cdot L \cdot N_{\text{layers}} \cdot N_{kv_heads} \cdot d_{\text{head}} \cdot b_{\text{prec}},$$

where the factor 2 accounts for separate K and V tensors and $b_{\text{prec}} = 2$ B for FP16.

(NVIDIA Developer) Instantiated for the reference models (all Grouped-Query Attention except Llama-2-7B):

Model	Ctx 4K (B=1 / B=32)	Ctx 32K	Ctx 128K	Ctx 1M
Llama-2-7B (MHA)	2.0 / 64.0 GiB	16 / 512 GiB	64 GiB / 2 TiB	512 GiB / 16 TiB
Llama-2-70B (GQA-8)	1.25 / 40 GiB	10 / 320 GiB	40 GiB / 1.25 TiB	320 GiB / 10 TiB
Llama-3-70B (GQA-8)	1.25 / 40 GiB	10 / 320 GiB	40 GiB / 1.25 TiB	320 GiB / 10 TiB

GQA buys a constant $8\times$ factor but does not break the linear scaling in $L \cdot B$. This is the regime KVQuant and KIVI explicitly target — Llama-2-70B at $128 K \times \text{batch } 32$ needs more KV bytes than an H100 has HBM. (arXiv) (UC Berkeley Statistics)

Why $K \neq V$ for quantization

The decode-phase attention decomposes as $A = \text{softmax}(q_t X_K^\top / \sqrt{d})$ then $o_t = A X_V$. **Keys feed an exponential; values feed a bounded linear combination.** Empirically (Liu et al., ICML 2024; Hooper et al., NeurIPS 2024), X_K exhibits *persistent channel-fixed outliers* — a handful of feature channels whose magnitude is $10\text{--}100\times$ the typical channel and stable across tokens — while X_V has a flat per-channel distribution. :antCitation[] {citations="fc62da3c-f032-4a3b-927a-630e9e1d328a"} Per-token (row-wise) K quantization therefore hands its entire dynamic range to a few outlier channels, leaving 2–3 effective INT8 levels for everything else; :antCitation[] {citations="44f9e934-ee1a-4ca7-ab7a-ed435df05779"} the softmax then amplifies the damage via exp. KIVI's ablation on Llama-2-13B shows CoQA accuracy collapsing from **66.37 (FP16) $\rightarrow$ 2.88** when K is quantized per-token at 2-bit, versus **63.53** with the asymmetric per-channel-K/per-token-V configuration. :antCitation[] {citations="fb6d9915-4250-4da2-ad91-58a138001dd7"} The universal principle distilled from this first-wave literature: **quantize along the axis that contains the outliers — K by channel, V by token.** :antCitation[] {citations="90b2beff-dff0-4886-af32-93f4b4e2ca05"}

Pre-RoPE versus post-RoPE: a live design axis

RoPE applies a position-dependent 2-D rotation R_m to K; each outlier channel is mixed with its pair and its apparent magnitude oscillates with $\sin(m\theta_i) / \cos(m\theta_i)$. Pre-RoPE per-channel K statistics are *stationary* in channel index; post-RoPE statistics widen and shift with position, invalidating offline calibration. :antCitation[] {citations="0c1d5d2a-f9ce-486a-9546-ffbced77a8b7"} KVQuant's ablation on LLaMA-7B shows :antCitation[] {citations="2954a44e-3707-4ae3-98ea-f2fd01f207bb"} **0.82 PPL improvement** from pre-RoPE alone at 3-bit. :antCitation[] {citations="b8786768-c45c-4924-ac79-9ce160108a65"} The cost of pre-RoPE storage is that attention must rotate K on-the-fly for every query via a fused RoPE-dequant kernel; :antCitation[] {citations="3c67311e-d8bb-47b3-95a8-a9ce884d03f0"} most production stacks (vLLM, TensorRT-LLM, llama.cpp) store post-RoPE K, which is why they cap at 3–4 bit and why QuaRot/SpinQuant/PolarQuant's *post-RoPE rotation* approach has become the dominant alternative path.

2. A taxonomy of SOTA quantization schemes

Four orthogonal axes organize the entire field.

Numeric format. Uniform schemes (INT4/INT8 RTN with affine $Q(x) = \lfloor (x - z)/s \rfloor$) trade simplicity for range efficiency; (arxiv) non-uniform codebooks (NF4, KVQuant's (nuqX) Fisher-sensitivity-weighted k -means, (arXiv) SqueezeLLM centroids) buy accuracy at the cost of lookup-table dequant. Block-floating-point formats now dominate hardware: **FP8 E4M3** ($1s+4e+3m$, ± 448 , Hopper/Blackwell Tensor Cores), **FP8 E5M2** ($1s+5e+2m$, wider range, lower precision), (vLLM) (NVIDIA) **MXFP4** (32-element block, FP4 E2M1 element, UE8M0 power-of-two scale $\rightarrow 4.25$

bpe) and **NVFP4** (16-element block, (NVIDIA Developer) UE4M3 scale $\rightarrow$ 4.5 bpe). (NVIDIA Developer) NVFP4's finer blocks and higher-precision scales consistently outperform MXFP4 (NVIDIA Developer +2) by ~ 0.25 bpe worth of accuracy, (NVIDIA Developer) and for KV cache MXFP4 is empirically inadequate without a Hadamard rotation because V-cache outliers overflow power-of-two scaling.

Granularity. Per-tensor is obsolete; per-token (row-wise) suits V; per-channel (column-wise) suits K but is streaming-incompatible without a residual buffer. Group-wise schemes ($g=32/64/128$) give effective bit-width $B_{\text{eff}} = B + (16 + B)/g$, e.g., INT4-g128 asymmetric = 4.16 bpe. **Smaller groups improve accuracy but compound dequant overhead**; Atom uses $g=128$, KIVI $g=32$, KVQuant per-vector.

PTQ versus QAT. KV quantization is almost exclusively post-training because KV activations are intermediate, not parameters, and simulating the full generation loop during QAT is prohibitive. LLM-QAT (Liu et al., 2023) remains the main exception; NVIDIA's Model Optimizer added experimental KV-QAT in 2025 but published results are still dominated by PTQ.

Static versus dynamic. Weight quantization is static; KV quantization is predominantly dynamic (online) with per-token scales computed at write time. KVQuant showed that offline-calibrated pre-RoPE per-channel K scales are viable when paired with outlier extraction (arXiv) (Berkeley) — essential because online channel statistics require a top-k over the channel dimension every step.

3. Algorithmic deep dive — the canonical methods

First wave: asymmetric quantization and outlier decomposition (2023–early 2024)

SmoothQuant (Xiao et al., ICML 2023, arXiv:2211.10438) introduced the outlier-migration transformation $Y = XW = (X \text{diag}(s)^{-1})(\text{diag}(s)W) = \tilde{X}\tilde{W}$, with per-channel scale $s_j = \max_i |X_{ij}|^\alpha / \max_k |W_{jk}|^{1-\alpha}$. Every subsequent KV method inherits this trick, and it reappears transformed as QServe's SmoothAttention (migrating K outliers into Q). (arXiv)

Atom (Zhao et al., MLSys 2024, arXiv:2310.19102) (University of Washington) was the first end-to-end W4A4KV4 pipeline. It reorders the 128 highest-energy channels into an INT8/FP16 slice, (MLsys) quantizes the rest at INT4-g128, and fuses dequant with MMA via INT32 partial-sum accumulation. (arXiv) Llama-7B W4A4KV4 WikiText-2 PPL rises only +0.48 over FP16; end-to-end **7.7× throughput over FP16 serving**. (GitHub) Atom's key weakness — per-token KV quantization with no K-per-channel treatment — is precisely what KIVI and KVQuant fix.

KIVI (Liu et al., ICML 2024, arXiv:2402.02750) is the reference **2-bit tuning-free PTQ**. (GitHub) Per-channel K + per-token V, (arXiv +3) group size 32, (arXiv) residual FP16 buffer (arxiv) of 128 recent tokens (GitHub) (arXiv) flushed into quantized groups. Llama-2-7B WikiText-2 PPL FP16 5.47 $\rightarrow$ KIVI-2 5.52–5.57; LongBench neutral; **2.6× peak memory reduction, 4× larger batch, 2.35×–3.47× throughput on A100**. (arXiv +4)

KVQuant (Hooper et al., NeurIPS 2024, arXiv:2401.18079) stacks five techniques — per-channel K, **pre-RoPE K**, non-uniform (nuqX) Fisher-weighted k -means, per-vector Dense-and-Sparse

with top-1 % FP16 outliers, (arXiv +5) and attention-sink-aware + Q-Norm (Hugging Face) — to drive 3-bit PPL within 0.07 of FP16 on Llama-7B. This is the method enabling **1 M context on a single A100-80GB and 10 M context on an 8-GPU server.** (NeurIPS +6)

WKVQuant (Yue et al., arXiv:2402.12065) demonstrates **Past-Only Quantization**: the just-computed (k_t, v_t) stay FP16 for the current attention and are quantized only when written to cache, (arXiv) (arXiv) removing one round of quant/dequant error on the highest-leverage token.

Method	Venue	Bit-width	K scheme	V scheme	RoPE	PPL Δ (L2-7B)	Speedup / Memory
SmoothQuant	ICML'23	W8A8	per-tensor (smoothed)	per-tensor	post	$\leq +0.1$	1.56× E2E, 2× mem
Atom	MLSys'24	W4A4KV4, g=128	per-token + 128-ch FP16 outliers	per-token	post	+0.12- 0.48	7.7× vs FP16
KIVI	ICML'24	KV2 / KV4, g=32, R=128	per-channel	per-token	post	+0.05- 0.10 @ 2-bit	2.35- 3.47× tput
KVQuant nuq3-1%	NeurIPS'24	3.32 bpe	per-channel pre-RoPE + 1% sparse	per-token + 1% sparse	pre	+0.07	4.8× KV; 1M ctx on A100
KVQuant nuq2-1%	NeurIPS'24	2.32 bpe	same + Q- Norm	same	pre	+0.5	6.9× KV
WKVQuant	arXiv'24	W4-KV4	2-D (channel × token)	2-D	post	$\approx$ GPTQ- W4	memory- only

Second wave: hardware-aware, saliency-aware, and decomposition methods (mid-2024-early 2025)

QServe / QoQ (Lin et al., MLSys 2025, arXiv:2405.04532) (MIT HAN Lab) is the systems-aware successor to Atom: **W4A8KV4** with progressive INT8→INT4 weight dequant (Semantic Scholar) inside the INT8 MMA, SmoothAttention migrating K outliers into Q offline, and register-level INT4→INT8 parallel decode. (arXiv) Llama-2-7B W4A8KV4 PPL 5.75 (+0.28); **1.2-1.4× A100 and L40S speedup** vs TensorRT-LLM on Llama-3-8B, **2.4-3.5×** on Qwen1.5-72B, (arXiv) (MIT HAN Lab) cutting \$/token by $\sim 3\times$.

GEAR (Kang et al., NeurIPS ENLSP 2024, arXiv:2403.05527) decomposes KV as $X \approx Q(X) + L + S$ where L is a rank- r SVD approximation of the residual and S holds the top 2 % sparse outliers. (arXiv) At 2-bit GEAR beats KIVI-2 by **+14.95 % average accuracy** on GSM8K/AQuA/BBH/LongBench; (arXiv) 2.38× throughput, 2.29× peak memory. (Hugging Face) (arXiv)

QAQ (Dong et al., arXiv:2403.04643) provides the first formal theorem that K-quant error is exponentially amplified by softmax whereas V-error is linearly averaged, (arXiv) and allocates extra bits to K accordingly; achieves up to **10× compression** (arXiv) (ResearchGate) with <1 pt PIQA loss on Llama-2-7B.

ZipCache (He et al., NeurIPS 2024, arXiv:2405.14256) introduces the **normalized attention score** — corrected for the fact that later tokens cannot attend to early ones (Openvino) — as a saliency metric, combined with channel-separable token-wise (Openvino) 4/2-bit mixed precision computed from a small probe of recent queries (so FlashAttention remains usable). Mistral-7B GSM8K at **4.98× compression with 0.38 % accuracy loss**; (Openvino) (arXiv) Llama-3-8B at 4 K context: prefill −37.3 %, decode −56.9 %, memory −19.8 %. (arXiv)

****KVTuner**** (Li et al., ICML 2025, arXiv:2502.04420) is the state of the art in **mixed-precision layer-wise search**, using intra-layer Pareto pruning + inter-layer clustering :antCitation[] {citations="ae7245fb-fd3b-468f-aa7a-7d8c4fffb050"} over pairs $(b_K^\ell, b_V^\ell) \in \{(8, 8), (8, 4), (4, 8), (4, 4), (4, 2), (2, 4), (2, 2)\}$. Achieves ****3.25-bit nearly lossless**** on Llama-3.1-8B-Instruct :antCitation[] {citations="e8e55d1f-3c56-4689-bc84-41f3e88e864a"} with +38.3 % throughput over KV8 :antCitation[] {citations="83ff545a-47d2-4bfc-beeb-2d1660552dbb"} KIVI.

IntactKV (Liu et al., ACL Findings 2024, arXiv:2403.01241) preserves the **pivot tokens** — first 1-4 BOS/sink tokens whose activations exceed $1e3$ in specific channels (arXiv) — at FP16 by generating them with the full-precision model (arXiv) and concatenating with the quantized remainder; pairs with AWQ/GPTQ/OmniQuant/QuaRot (arXiv) to close 30-50 % of their residual quant gap.

MiKV (Yang et al., ICML 2024, arXiv:2402.18096) replaces H2O-style eviction with *demotion* to INT2/INT4 so no token's information is destroyed, plus channel smoothing on Q and K. Up to 80 % compression; (arXiv) later surpassed on Line-Retrieval by ZipCache (+18.27 %) (NeurIPS) thanks to the latter's normalized-saliency metric.

Q-Hitter (MLSys 2024) unifies heavy-hitter selection with a per-layer **quantization-friendliness metric** (MLsys) (inverse L_∞ norm of the K/V vector), yielding tokens that are both important and compressible: **20× KV memory reduction, 33× throughput vs HF Accelerate**.

(Eindhoven University of Te...)

Third wave: rotations, polar coordinates, vector quantization, sub-2-bit (late-2024-April 2026)

QuaRot (Ashkboos et al., NeurIPS 2024, arXiv:2404.00456) fuses four Hadamard rotations (R_1, R_2 offline into weights (Amd) via SliceGPT's computational invariance; R_3, R_4 online) (Amd)

so activations, keys, and values become incoherent. R_3 is applied **post-RoPE** to the Q/K pair (Amd) so attention is unchanged. (arXiv) Llama-2-70B W4A4KV4 WikiText-2 PPL 3.79 vs 3.32 FP16 (alphaXiv) ($\Delta \leq 0.47$), retaining ~99 % zero-shot accuracy. (arXiv) This established the post-RoPE Hadamard path that has since become the dominant way to enable 4-bit KV in production frameworks.

SpinQuant (Liu et al., ICLR 2025, arXiv:2405.16406) replaces QuaRot's random Hadamards with **learned rotations on the Stiefel manifold via Cayley SGD**, (arXiv) (Emergent Mind) optimizing the end-to-end quantized loss. (arXiv) Llama-2-7B W4A4KV4 zero-shot avg 64.0 (2.9 pt gap to FP16, beating LLM-QAT by 19.1 pts); (arXiv +2) closes **45.1 % of the residual QuaRot gap on the harder-to-quantize Llama-3-8B**. (ADS) (arXiv)

SKVQ (Duanmu et al., COLM 2024, arXiv:2405.06219) permutes KV channels so groups contain statistically similar channels, applies clipped dynamic quantization, (arXiv) and retains a sliding window plus attention sinks in FP16. (ADS +3) Pushes to **K2V1.5** (2-bit keys, 1.5-bit values) (arXiv) with <2 pt LongBench loss (arXiv) on LLaMA-2-7B-80K and LongChat-32K; **1 M context on a single A100, ~7× decode speedup**. (arXiv) (ADS) Uses pre-RoPE key quantization.

Palu (Chang et al., ICLR 2025, arXiv:2407.21118) caches only a **low-rank latent** $H = XA$ from an SVD-decomposition $W_K = A_k B_k$, $W_V = A_v B_v$, reconstructing K,V on the fly via a fused kernel; (arXiv +2) absorbs B_v into W_O . (arXiv) 50 % KV compression with ≤ 1 PPL loss (arXiv) (PromptLayer) and 1.89× attention speedup (OpenReview) (2.91× combined with 3-bit quantization). (OpenReview) (arXiv)

PQCache (Zhang et al., SIGMOD 2025, arXiv:2407.12820) reframes KV management as ANN retrieval: CPU-side k-means builds **product-quantization codebooks** over post-RoPE keys, (arXiv) and decoding retrieves top- k relevant tokens (arXiv) for exact attention. (arXiv) 75 % KV memory cut with ≥ 97 % FP16 quality (Zerna) on LongBench; (GitHub) **+4.6 pts over prior methods on InfiniteBench**. (arXiv) (arXiv)

MiniCache (Liu et al., NeurIPS 2024, arXiv:2405.14366) observes very high KV cosine similarity between adjacent middle-to-deep layers (arXiv) and **SLERP-merges directions** while retaining magnitudes (Hugging Face) plus an unmergable-outlier set. (Hugging Face +2) On Llama-3-70B, merging 87.5 % of layers gives near-zero COQA drop; (arXiv) (arXiv) stacked with 4-bit quant yields **5.02× KV reduction**. (arXiv) (The Moonlight)

KVSharer (Yang et al., ICLR 2025, arXiv:2410.18517) exploits the counter-intuitive finding that **dissimilar** donor layers share KV best; (arXiv +2) 70 % memory at >95 % accuracy, (arXiv) 1.3× generation speed, (ADS) (arXiv) composable with KIVI (arXiv) for multiplicative gains.

PolarQuant (two independent concurrent papers, Wu et al. arXiv:2502.00527 and Han et al. arXiv:2502.02617) exploits the 2-D rotational structure of RoPE: represent each pair (k_{2i}, k_{2i+1}) as (r, θ) ; outliers are confined to one coordinate, yielding well-structured radius and angle distributions (arXiv +2) quantized via per-channel codebooks. Attention becomes a table-lookup $\langle q, k \rangle$. (ResearchGate +3) SOTA on NIAH 4 K-104 K on Llama-3.1-8B-Chat. (ResearchGate) (Help Net Security)

CommVQ (Apple, ICML 2025, arXiv:2506.18879) trains **RoPE-commutative additive vector-quantization codebooks** (Apple Machine Learning R...) via EM so the codebook integrates into self-attention. (Semantic Scholar) 2-bit gives 87.5 % KV reduction (near-lossless); **1-bit is viable**, enabling LLaMA-3.1-8B at 128 K context on a single RTX 4090. (OpenReview)

TurboQuant / QJL (Zandieh et al., Google 2026) applies a **1-bit Johnson-Lindenstrauss sign hash** to keys (GitHub) with an unbiased inner-product estimator, eliminating scale/zero-point overhead entirely. Matches FP16 at 3.5-bit equivalent on LongBench (Li (EMSI) ama-3.1-8B: 50.06 vs 50.06) (TechInformed) with **8× H100 attention-logit speedup**; (Help Net Security) values still need ≥2 bits.

H1B-KV (arXiv:2510.05529) combines 1-bit key sign sketches with 4-bit values for a **70× footprint reduction** — 60 MB total cache for 7B at 8K context — matching full precision on GSM8K/MMLU/HumanEval after light fine-tuning.

XQuant (arXiv:2510.11236) stacks data-free calibration with post-quant cross-layer KV sharing to reach **<1.4 bits effective**, beating KIVI-2 and AsymKV-1.5 (arXiv) on LongBench (arXiv) (ACL Anthology) (Mistral-7B, Qwen2.5-14B).

Other notable 2025 entries include **AnTKV** (anchor-token VQ, 0.375–1 bpe, 810 K context on single A100), **NSNQuant** (calibration-free VQ via Normalize-Shift-Normalize + Hadamard, NeurIPS 2025), **VecInfer** (smooth+Hadamard VQ, 2.7× attention / 8.3× end-to-end), **MILLION** (outlier-immunized PQ), **MiniKV** (ACL 2025 Findings, pyramid eviction + 2-bit sub-channel), **Squat** (subspace-orthogonal), **PM-KVQ** (progressive block-wise for long-CoT reasoning), and **RCP** (W2A4KV4). The frontier's bit-width ceiling has moved from 4-bit (early-2024: QuaRot/SpinQuant) to 2-bit (late-2024: KIVI, SKVQ) to **sub-2-bit (2025-2026: CommVQ 1-bit, AnTKV 0.375-bit, XQuant <1.4-bit)** — almost entirely via vector quantization and cross-layer sharing rather than scalar refinements.

4. Hardware and systems integration

Tensor-core-native formats as the binding hardware constraint

The generational sweep of GPU hardware determines what is actually deployable:

GPU	HBM BW	Native low-bit TC formats
A100 (Ampere)	2.0 TB/s	INT8, INT4 (no FP8 hardware)
H100 / H200 (Hopper)	3.35 / 4.8 TB/s	FP8 E4M3/E5M2 via Transformer Engine; 3,958 TFLOPS FP8
B100 / B200 (Blackwell)	8.0 TB/s	NVFP4, MXFP4, FP6, FP8; 5th-gen TC; ~2.3× H100 FP8 TFLOPS

Because decode is memory-bound, going **H100 → H200 → B200 alone delivers 1.43× → 2.4× decode speedup** for unchanged kernels — quantization multiplies on top of this. This is why

Semianalysis and NVIDIA call H200 the "inference miracle": it is mostly KV-bandwidth arbitrage. NVFP4's 16-element block with UE4M3 scale (note: *unsigned* E4M3 — MLX currently ships a buggy signed variant) strictly outperforms MXFP4's 32-element block with UE8M0 power-of-two scale for KV because V-cache outliers overflow power-of-two block scaling.

FlashAttention-3 and the fused-dequant gospel

FlashAttention-3 (Shah et al., NeurIPS 2024, arXiv:2407.08608) is the reference design. It reaches **740 TFLOPS (75 % utilization)** in FP16 and **~1.2 PFLOPS in FP8** on H100 via warp-specialization, GEMM-softmax interleaving, and hardware FP8 paths with incoherent (Hadamard) processing. Crucially, FA3's FP8 path requires **Q also in FP8**: vLLM's FA3 backend dynamically quantizes queries per-row so the PV GEMM stays on FP8 Tensor Cores rather than dequantizing K/V before the dot-product. This is the difference between **storage-only FP8 KV** (vLLM's pre-FA3 mode, which doubled capacity without improving latency because dequant-then-attend was unfused) and **compute-in-FP8 attention** (FA3, TensorRT-LLM's FP8 attention plugin).

For sub-8-bit KV, on-the-fly dequant must live inside the attention inner loop. QServe measured naive KV4 attention as $1.7\times$ faster than TRT-LLM-KV8 on L40S but **1.1–1.2× slower on A100** because the extra dequant CUDA-core ops pushed the operator into the compute-bound region. The canonical INT4→FP16 dequant trick, inherited from Marlin (IST-DASLab) and FasterTransformer's (`interleaved_numeric_conversion.h`), uses a single (`lop3.b32`) PTX instruction to unpack two INT4 values per INT32 register by ANDing with (`0x000f000f`)/(`0x00f000f0`), ORing with (`0x64006400`), reinterpreting as FP16 (`1024 + x/16`), and subtracting — two FP16 values per INT32 per instruction, perfectly shaped for (`mma.sync`). Sparse-Marlin extends this to 2:4 sparsity (INT4 + 2:4) for another $\sim 2\times$ boost.

PagedAttention, vLLM, and the serving framework zoo

vLLM's PagedAttention stores K/V in 16-token blocks referenced by per-sequence block tables; quantization is applied per-block via (`reshape_and_cache`) with per-tensor scalars (`k_scale`, `v_scale`) loaded from llm-compressor calibration JSON. Current (April 2026) framework support:

Framework	KV formats	Attention fusion	Representative result
vLLM ≥0.9	FP8 E4M3 (default), E5M2, FP4 E2M1 (experimental on Blackwell)	FA3: Q+K+V fused FP8; default PagedAttn: dequant→FP16 (capacity only)	INT8 KV removed in favor of FP8
TensorRT-LLM	FP8 E4M3, INT8, NVFP4	Native FP8 attention plugin; NVFP4 KV via ModelOpt	FP8 KV: 1.09× prefill, 1.45× decode vs vLLM (SqueezeBits)
SGLang	FP8, FP4 (experimental)	Fused dequant with FA backend; loud warning if backend lacks fusion	—
LMDeploy (TurboMind)	INT8, INT4 KV online per-head per-token asymmetric	Custom TurboMind attention with fused dequant	W4A16+KV4: +40 % RPS vs FP16 on Llama2-7B
llama.cpp (GGUF)	Q8_0, Q4_0 K/V independent ($\frac{\text{cvt}(\text{rnsatfinite.e4m3x2.f16x2})}{\text{ctk}} \big/ \frac{\text{cvt}(\text{rnsatfinite.e4m3x2.f16x2})}{\text{ctv}}$)	FA CUDA path ($\frac{\text{cvt}(\text{rnsatfinite.e4m3x2.f16x2})}{\text{ctk}} \big/ \frac{\text{cvt}(\text{rnsatfinite.e4m3x2.f16x2})}{\text{ctv}}$) fuses dequant	Q4_0 K+V: 72 % KV ↓ but −37 % decode at 110K ctx on consumer GPUs lacking efficient INT4 paths
QServe/omniserve	KV4 + W4A8 (QoQ)	Custom fused KV4 attention	2.36× geomean vs TRT-LLM 0.9.0 on L40S, 1.68× on A100
TRT-LLM Blackwell	NVFP4 KV (block-16, E4M3 scale)	New NVFP4 attention kernels via CUTLASS 3.x	Paired with FP8 weights; requires ModelOpt

The headline insight across these numbers: **fusion quality determines whether quantization is free or costly**. vLLM's pre-FA3 FP8 KV actually slowed prefill slightly; llama.cpp's Q4_0 lost 37 % decode throughput at 110 K context on DGX Spark GB10 despite the 72 % memory savings. In contrast, QServe's aggressive per-channel-K-fold-into-Q and per-token-V-scale-into-softmax fusion makes 4-bit KV the faster path even at batch 1 on H100.

FP8 and NVFP4 in production

Hopper's E4M3 (range ±448, preferred for KV) and E5M2 (wider range, used before calibration tooling matured) are produced by `cvt.rn.satfinite.e4m3x2.f16x2` PTX, consumed by `wgmma.mma_async` with FP32 accumulate. Calibration flows (llm-compressor for vLLM, NVIDIA ModelOpt for TRT-LLM) run ~512-sample passes; dynamic-scale modes cost 1–2 % throughput.

Blackwell's NVFP4 is consumed directly by the `tcgen05.mma` instruction with `mx4`/`nv4` variants — the block scale is applied inside the MMA with no explicit multiply. NVIDIA's

December 2025 blog reports **NVFP4 KV at ~50 % of FP8 memory, <1 % accuracy loss** on MMLU-PRO, MBPP, LiveCodeBench, and RULER-64K for Qwen3-480B-A35B, with up to 3× TTFT improvement via prefix-cache hit-rate gains. Llama-3.3-70B-NVFP4, DeepSeek-R1-NVFP4, and DeepSeek-V3.2-NVFP4 ship publicly. Crucially, **for DeepSeek-V3.2's MLA, NVIDIA keeps weights at NVFP4 but KV at FP8** — a strong signal that 4-bit KV on MLA is not yet production-stable. No fully-fused NVFP4 attention kernel on par with FA3-FP8 exists in public vLLM/SGLang as of April 2026; CUTLASS 3.x has the primitives but parity is expected only by mid-2026.

5. Open problems and the next frontier

Sub-2-bit robustness under long-CoT reasoning remains unsolved. KIVI at 2-bit loses 10–15 pts average on Qwen3-8B and Llama-3-8B reasoning tasks (per KITTY, arXiv:2511.18643); AIME-24/25 and LiveCodeBench expose error accumulation invisible to WikiText-2 PPL. Progressive-bitwidth schemes (PM-KVQ) and vector-codebook approaches (CommVQ, H1B-KV, TurboQuant) are the two leading paths, but none has shown consistency across RULER-128K, NeedleBench-V2 MultiNeedle-128K, and Text2JSON (arXiv:2604.08426) simultaneously.

Interaction with architectural KV reduction is non-commutative. GQA and especially DeepSeek's **Multi-head Latent Attention** already compress cache 4–8× or more, leaving less redundancy for quantization. MLA compresses via a joint low-rank projection $c_t^{KV} = W^{DKV} x_t$ (V2: $d_c = 512$) with a **decoupled RoPE sub-head** — each query head gets a small RoPE-carrying sub-head plus a single shared MQA-style RoPE key, because standard RoPE does not commute with the up-projection. This is what makes $\text{MLA} \times \text{quantization}$ subtle: the non-RoPE latent is easy to quantize low, but the small RoPE sub-head still carries RoPE-induced outliers and must stay high-precision. NVIDIA's production DeepSeek-V3.2 deployment confirms 4-bit MLA KV is still unreliable. DeepSeek-V3 additionally trains natively in **FP8 E4M3** with fine-grained 1×128 tile / 128×128 block scaling, making it the first flagship model where FP8 is the pretraining precision.

Eviction, sparsity, and quantization compound errors in unpredictable ways. Heavy-hitter eviction (H2O, SnapKV, PyramidKV) concentrates quant error into retained salient tokens; Q-Hitter's combined oracle $S = \alpha \cdot \text{AttnScore} - \beta \cdot \text{QuantErr}$ is the cleanest principled combination. Dynamic retrieval sparsity (Quest, DoubleSparse, ShadowKV) depends on MIPS recall that quantized keys degrade; ShadowKV keeps pre-RoPE low-rank keys in FP16 on GPU while offloading values. Composing **MLA → MiniCache cross-layer merge → KVTuner mixed-precision quant → PQCache retrieval-offload → LMCACHE CXL tier** is the emerging production stack, but the cumulative error budget is under-characterized.

Competing non-quantization KV compression has matured rapidly. Dynamic Memory Compression (Piwowarski et al., ICML 2024) retrofits learnable per-token merging $\alpha_t \in \{0, 1\}$ via continued pretraining for up to **4× compression and 3.7× H100 throughput**, beating GQA. Dynamic Memory Sparsification (NeurIPS 2025, arXiv:2506.05345) achieves 8× compression via a 1K-step retrofit delayed-Gumbel-sigmoid eviction, **+9–12 pts on Qwen-R1-32B AIME24** at equal memory reads. These methods are orthogonal to quantization and eventually must compose with it.

Evaluation protocols are the current bottleneck to progress. WikiText-2 PPL is insensitive beyond 4 bits; the community has moved to RULER's 13-task suite at 4K-1M, LongBench-v2, ∞ Bench, and agentic/multi-turn evaluations (arXiv:2601.09527 benchmarks 79 configs on RAG, multi-LoRA, high-concurrency APIs). Reasoning benchmarks (AIME, GPQA-Diamond, LiveCodeBench) are where sub-4-bit methods now expose their real costs.

Hardware codesign is the most exciting short-term axis. The combination of NVFP4 Tensor Cores + CUTLASS 3.x + fused FlashAttention-3-style kernels will likely make 4-bit KV the default on Blackwell within a year; the open questions are (i) robust KV quant algorithms **native** to the NVFP4 block layout rather than emulating INT4; (ii) how to expose mixed-bit KV to **PIM** fabrics (Samsung HBM-PIM, SK Hynix AiM) that naturally fit attention GEMV; and (iii) whether **CXL/NVMe-resident quantized prefix caches** (LMCache arXiv:2510.09665, Mooncake, 3FS, NVIDIA Dynamo) can avoid requantization drift when CacheBlend-style prefix composition is applied. Research accelerators like Oaken (ISCA 2025) and BitDecoding push further with fused dense-sparse outlier encoding.

The most underexplored frontiers. Speculative decoding with quantized KV (QuantSpec, arXiv:2502.10424) shares hierarchical 4-bit KV between target and draft via bit-slicing with preserved acceptance rates — a particularly clean co-design. Diffusion LLMs (LLaDA, Fast-dLLM) break causal KV caching and have almost no published quantization literature. Persistent prefix caching for multi-agent workloads (22-136 $\times$ TTFT wins on edge) will force quantization algorithms to survive *requantization* upon prefix recomposition — today's drift accumulates fast enough to break any sub-4-bit scheme under realistic agentic traffic. Finally, KV-aware QAT is now supported in NVIDIA's Model Optimizer but has essentially no published results; this is the most likely path to robust sub-2-bit KV on NVFP4 hardware.

Conclusion — where the field has actually moved

Three shifts over 30 months define the current state of the art. **First**, the K-per-channel / V-per-token asymmetry is universally accepted, theoretically grounded, and implemented in every production framework either directly (LMDeploy, KIVI-style) or indirectly via pre-rotation that makes per-token-everything feasible (QuaRot, SpinQuant, PolarQuant). **Second**, the bit-width ceiling has dropped from 4-bit to sub-2-bit, but only via **non-scalar** methods — vector quantization, product quantization, low-rank plus sparse decomposition, cross-layer sharing, and 1-bit JL sign hashing — because scalar sub-2-bit fundamentally cannot capture the long-tailed distributions of post-RoPE keys without codebooks or dimensional coupling. **Third**, hardware-software co-design has decisively moved from INT4 emulation to block-floating-point formats (FP8 E4M3, NVFP4), and the binding constraint is now the maturity of fused attention kernels, not the quantization algorithm itself. The field is converging on an architectural answer — MLA and DMC compress before caching, quantization compresses what's left, and CXL/PIM handles what the HBM cannot hold — but the error budgets of that stack and its behavior under long-CoT reasoning and agentic workloads remain the open research frontier through 2026.